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Evaluation and inter-comparison of weighing gauge measurements in Basque Country.

Santiago Gaztelumendi[1]; Kepa Otxoa de Alda[1]; José Antonio Aranda[1]

[1] Basque Meteorology Agency (Euskalmet)

Corresponding author: Santiago Gaztelumendi, s-gaztelumendi@euskalmet.eus

Abstract:

A comparative analysis was conducted using six different rainfall gauge datasets located at the Arkauti site in the Basque Country. Rainfall data, collected at one-minute intervals from five distinct weighing gauges (2 Geonica, 2 OTT, 1 Lambrecht), and data from an operational tipping-bucket rain gauge (Geonica) at 10 minutes intervals, located at the same site, are prepared for a continuous five-month period and two isolated episodes.

The study has two main objectives. First, to evaluate and compare the performance of five different in-situ weighing rain gauges against each other and relative to the currently operational tipping-bucket rain gauge. Second, to establish a robust methodology and develop a set of scripts to facilitate the comparative study of rainfall series, allowing for repeated application in similar scenarios in the future—either when longer data series become available or when applied to other studies.

This contribution details the data selection and preparation process, the methodology used for evaluation, inter-comparison, and performance analysis, along with key results, discussions, and conclusions. The ultimate goal is to enhance our understanding and expertise, supporting future decisions regarding precipitation data acquisition as part of the ongoing modernization of the Basque AWS network.

1. Introduction

Since its creation in 1990, the Basque Service of Meteorology (BSM) has been a reference in Basque Country for daily demand of meteorological data (e.g. GV 1990, 1997b, 1999b). At present Basque Meteorology Agency (Euskalmet) remain at front end of operational meteorological affairs for Basque Country, including the maintenance and improvement of the Basque Automatic Weather Stations (AWS) network (e.g. Euskalmet 2022, Gaztelumendi et al 2018).

The AWS network, which began in the late 1980s with 20 stations, has evolved over the last 30 years through better instrumentation and increased density, now covering the entire territory (Gaztelumendi et al. 2018). Currently, the Basque Country Mesonet records more than 160,000 observations daily from over 120 AWS of various types, reporting every ten minutes without interruption. Campbell Scientific CR1000 and CR6 dataloggers are used. Regarding rain gauges, 96 AWS have a Geonica PCP-215 tipping bucket rain gauge installed (see Fig. 2). These rain gauges have been modified with additional features, such as optimized heating systems for recording all types of precipitation or totalizer deposits for total precipitation volume control. Besides the usual parameters available in these rain gauges, they also record the exact second of each tipping and provide information on heater operation (Gaztelumendi et al. 2018, 2022a, 2022b).

To gain expertise in the operation of other types of rain gauges, different from tipping-bucket types, a set of different gauges was installed at an experimental site to develop expertise in their operational use (Gaztelumendi et al. 2022a). This study aims to continue the path marked by an initial sensor comparison experiment (Gaztelumendi et al. 2022a), expanding its scope and addressing methodologies for inter-comparison of results, extracting conclusions about the strengths and weaknesses of each instrument in an operational context.

2. Methods and data

At the Arkauti study site, five different weighing rain gauges from Geonica, OTT, and Lambrecht were installed (see Fig. 1). The Arkauti site also has an operational rain gauge integrated into the mesonet, similar to the vast majority of gauges present in the network, that is a Geonica PCP-215 tipping bucket rain gauge. The experimental rain gauges were installed at Arkauti in late 2020, with preliminary data acquisition and transfer configurations set for one-minute data latency. However, due to some communication and configuration problems, data from the first stage were either missing or of poor quality (see Gaztelumendi et al. 2022a for details).

#	Manufacturer	Model	Type	Surface (cm2)	Data Latency
1	GEONICA	PCP-215	tipping-bucket	400	10-min
2	GEONICA	TRW415 (WP475)	weighing	400	1-min
3	GEONICA	DATARAIN 4000	weighing	300	1-min
4	OTT	PLUVIO2	weighing	400	1-min
5	OTT	PLUVIO2L	weighing	400	1-min
6	LAMBRECHT	RAIN[E]400	weighing	400	1-min

Figure 1: Rain-gauges in experimental Arkauti site.

For this study, a rigorous quality check and data curation were conducted to ensure a subset of data of sufficient quality for drawing reliable conclusions. The selected dataset includes a continuous five-month period (Dat03) from February 1, 2023, to July 1, 2023, as well as two isolated episodes: Dat01, spanning from September 29 to October 1, 2022, and Dat02, covering the same dates (see Fig. 2). The datasets (Dat01, Dat02, and Dat03) comprise 2,881, 4,321, and 216,001 one-minute records, respectively, corresponding to six different instruments: PGPCP, PGTRW, PG4000, POTT1, POTT2, and PLAMB. It is important to note that the official tipping-bucket rain gauge (PGPCP) originally records data at 10-minute intervals, so it was extrapolated to one-minute intervals to facilitate the analysis. All three datasets are analyzed together throughout much of this document.

The methodology used is based on two stages:

First Stage: Evaluation of Series. This stage involves statistical summaries, counting and analyzing rainfall events, and characterizing temporal aspects. Here, the most characteristic features of each series are evaluated in a comparative manner.

Second Stage: Intercomparison of Series. This stage employs various techniques, including binary analysis, maximum cross-correlation/lag analysis, and different metrics for assessing agreement, dissimilarity, and other measures of quantitative accuracy.

The authors have prepared several R scripts for data extraction, formatting, cleaning, calculation of derived quantities, and tabular, graphical, and statistical data analysis.

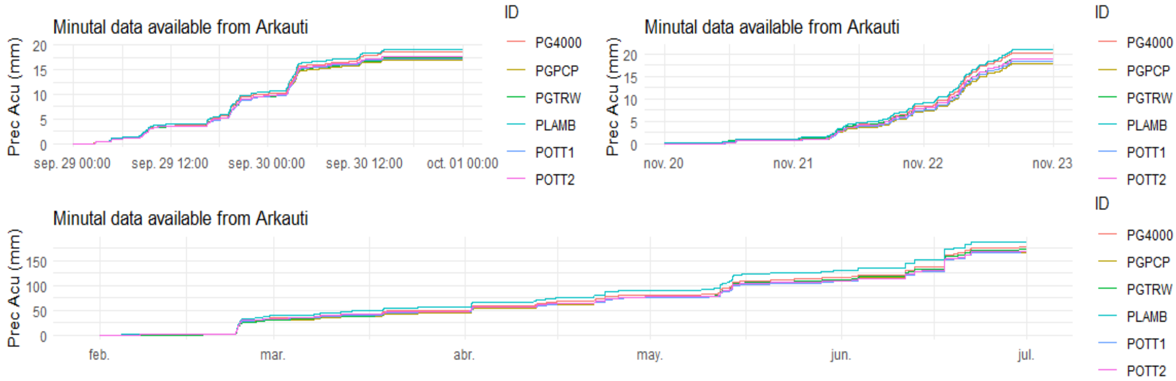

Figure 2: Acumulated precipitation in Dat01, Dat02 and Dat03 datasets

3. Results and discussion

Evaluation: Basic summary statistics.

In this initial evaluation step, basic summary statistics for non-zero events across different data series were analyzed. Table 1 shows that POTT2 has the highest mean and median for non-zero precipitation (0.09 and 0.06, respectively), indicating a tendency to record stronger rainfall events compared to other stations. In contrast, PGTRW and PLAMB have lower medians and means, suggesting a capacity to record lighter rainfall events. PLAMB has the highest resolution with a recording of 0.0007 mm/min, whereas POTT2 has the lowest resolution at 0.05 mm/10 min. Coefficient of Variation (CV) values exceeding 100% indicate that the standard deviation is larger than the mean, implying a high level of dispersion relative to the mean. This dispersion is more pronounced in PGTRW and less so in POTT2.

Table 1: Precipitation data sets summary statistics

Series	Mean	Median	Min	Max	Perc_95	Perc_98	SD	IQR	CV
PGPCP	0,03	0,01	0,0100	1,28	0,06	0,11	0,06	0,02	211
PGTRW	0,02	0,01	0,0010	1,85	0,06	0,11	0,06	0,02	305
PG4000	0,03	0,01	0,0033	1,83	0,09	0,13	0,07	0,01	222
POTT1	0,06	0,03	0,0100	1,87	0,14	0,23	0,09	0,04	171
POTT2	0,09	0,06	0,0500	1,83	0,21	0,32	0,12	0,04	124
PLAMB	0,02	0,01	0,0007	1,73	0,08	0,12	0,06	0,02	247

Mean / Median : Mean and median values for non-zero precipitation events.
Min / Max : Minimum and maximum values for non-zero precipitation events.
Percentiles 95/98 Total: Precipitation below which 95% and 98% of the observations fall.
SD NonZero: Standard deviation of non-zero events, indicating variability in rain intensity.
IQR: Interquartile range (IQR) reflects the range of the middle 50% of the data.
CV: Coeficient of variationvariation measure the relative dispersion, expresed in %.

Evaluation: Counts and analysis of pluviometry

The second evaluation step involves counting precipitation and non-precipitation events and checking the total accumulated amounts, means, maximums, and minimums. The analysis also considers counts of precipitation and non-precipitation periods appearing in each series, defined as the number of consecutive one-minute events with or without rainfall. Table 2 shows that the PLAMB records have the highest total accumulated precipitation (227.9 mm), while POTT2 shows the fewest rain events (2,195), suggesting it is less sensitive to detecting light rain. PGTRW shows a relatively high number of rain events (10,728) but a low average precipitation per event (0.13 mm), indicating a higher capability to register frequent, light rainfall.

Table 2: Counts and Accumulated Precipitation Table

Series	Rain Events	NoRain Events	NoZero Periods	Zero Periods	Total Accumulated Precipitation	Avg Accumulated Precipitation	Max Accumulated Precipitation	Min Accumulated Precipitation
PGPCP	7660	215543	302	303	201,1	0,66	19,9	0,1
PGTRW	10728	212475	899	862	208,9	0,13	21,525	0,001
PG4000	6642	216561	632	980	216,1	0,11	21,1836	0,003
POTT1	3659	219544	501	815	202,7	0,15	20,03	0,02
POTT2	2195	221008	314	892	204,7	0,16	20,16	0,05
PLAMB	9442	213761	713	1684	227,9	0,09	18,04615	0,0008

Rain Events: Number of periods with recorded precipitation.
NoRain Events: Number of periods with zero recorded precipitation.
NoZero Periods: Number of consecutive periods where precipitation was non-zero.
Zero Periods: Number of consecutive periods with zero precipitation.
Total Accumulated Precipitation: Sum of all recorded precipitation values.
Avg Accumulated Precipitation: Average precipitation recorded over all periods.
Max/Min Accumulated Precipitation: Maximum and minimum precipitation recorded in a single period.

Evaluation: Temporal chraracteristics.

The third evaluation step focuses on temporal characteristics, particularly the average, minimum, and maximum durations of rain and non-rain periods, and lead-lag times for each event. Lead and lag times provide insight into the number of events where a time series leads or lags precipitation detection relative to PGPCP. Table 3 shows that PGTRW has the longest average rain duration (7.0 minutes), indicating prolonged rainfall events (note that PGPCP1 min is artificially constructed from 10 min data). PGTRW also shows a significant number of lead events (4,186), suggesting it frequently detects precipitation before PGPCP does. POTT2 has higher lag counts, indicating a greater delay relative to PGPCP.

Table 3: Temporal Characteristics Table

Series	Avg Rain Duration	Max Rain Duration	Min Rain Duration	Avg NonRain Duration	Max NonRain Duration	Min NonRain Duration	Lead Count PGPCP	Lag Count PGPCP
PGPCP	25,4	210	10	711	21590	10		
PGTRW	7,0	271	1	139	21612	1	4186	1205
PG4000	3,7	131	1	119	13298	1	1581	2598
POTT1	2,8	44	1	167	13305	1	429	4541
POTT2	1,7	21	1	168	14375	1	225	5742
PLAMB	3,7	165	1	83	7018	1	3580	1798

Avg Rain Duration: Average duration of rain periods.
Max/Min Rain Duration: Maximum and minimum recorded duration of rain periods.
Avg NonRain Duration: Average duration of dry periods.
Max/Min NonRain Duration: Maximum and minimum recorded duration of dry periods.
Lead Count PGPCP / Lag Count PGPCP: Lead and lag counts in relation to the PGPCP series.

From the complete evaluation process, it can be concluded that there is considerable variability among the stations regarding rainfall frequency, intensity, and duration. Stations like POTT2 experience fewer but more intense rainfall events, while others, like PGTRW and PLAMB, record more frequent, lighter rainfall. Regarding temporal aspects, stations such as PGTRW show longer continuous rainfall durations, while pluviometers like POTT2 record very short rain durations. Lead and lag relationships suggest some instruments may be more sensitive to early rain events, such as PGTRW, which often detects precipitation before PGPCP.

Intercomparison : Binary analysis against PGPCP

We begin the intercomparison stage with what we can term a "binary analysis." This methodological approach is similar to those used when comparing forecasts versus observations. In our context, it provides a framework for evaluating the performance of the pluviometers relative to the "truth" of precipitation. Since this "truth" is not definitively known, we base our analysis on the "former known truth," which is the operational PGPCP data recorded at a 10-minute interval that has been used for a long time. For this purpose, all datasets are aggregated to 10-minute intervals and categorized into binary events to construct contingency tables and derive various metrics (see Table XX).

Different thresholds (e.g., >0, >1 mm) are used to define rain events, allowing us to analyze how different rainfall intensities affect the performance of the pluviometers. By analyzing the data in a binary format and calculating multiple evaluation metrics, we can examine the differential behavior of the pluviometers. Although this method is plausible for any aggregation time with any "truth," here we demonstrate its application to 10-minute aggregation using PGPCP as the "truth."

Table 4 presents an example of this analysis. The 10-minute dataset was filtered to include only events where at least one rain gauge registered some precipitation, and various thresholds (>0, >0.1, >0.3 mm) were considered to evaluate accuracy, sensitivity, specificity, and other relevant metrics. For the >0 threshold, the rain gauges exhibit a wide range of performance. PGTRW and PLAMB have relatively higher sensitivities (0.85 and 0.76, respectively), indicating that they are better at correctly identifying precipitation events. However, they have lower specificities (0.37 and 0.47), resulting in a higher rate of false positives. Conversely, POTT1 and POTT2 show high specificity (0.93 and 0.96), meaning they are more accurate in detecting true negatives but have lower sensitivities (0.41 and 0.25). These variations suggest that POTT1 and POTT2 are more conservative in detecting precipitation events, while PGTRW and PLAMB are more likely to detect precipitation but at the expense of accuracy in distinguishing between true and false positives.

When higher thresholds of >0.1 and >0.3 mm are considered, all gauges show marked improvements in specificity and precision, with values nearing 1. However, sensitivity decreases for most gauges, indicating they are less likely to detect true positive precipitation events as the threshold increases. At the >0.1 threshold, PG4000 and PLAMB demonstrate a good balance between sensitivity (0.73 and 0.66) and specificity (0.989 and 0.989), making them suitable in situations where minimizing both false positives and false negatives is equally important. However, POTT2 still shows significantly lower sensitivity (0.6), indicating a more conservative approach even at higher thresholds.

Table 4: Binary analysis metrics for 10-min datasets.

		Accu	Sens	Spec	PPV	NPV	F1	F1w	pTP	pTN	pFP	pFN	Kappa
>0	PGTRW	0,63	0,85	0,37	0,61	0,68	0,71	0,62	45,40	17,20	29,40	8,00	0,57
	PG4000	0,71	0,66	0,77	0,76	0,66	0,71	0,74	35,40	35,60	11,00	18,10	0,59
	POTT1	0,65	0,42	0,93	0,87	0,58	0,56	0,83	22,20	43,20	3,40	31,30	0,43
	POTT2	0,58	0,25	0,96	0,89	0,53	0,40	0,88	13,60	44,80	1,70	39,90	0,24
	PLAMB	0,63	0,77	0,47	0,62	0,64	0,69	0,61	41,00	21,70	24,90	12,40	0,54
>0.1	PGTRW	0,99	0,68	0,99	0,46	1,00	0,55	0,98	0,80	98,00	0,90	0,40	0,99
	PG4000	0,99	0,73	0,99	0,44	1,00	0,55	0,98	0,80	97,80	1,10	0,30	0,99
	POTT1	0,98	0,64	0,98	0,32	1,00	0,43	0,97	0,70	97,30	1,50	0,40	0,98
	POTT2	0,97	0,60	0,98	0,21	1,00	0,31	0,95	0,70	96,40	2,50	0,40	0,97
	PLAMB	0,99	0,66	0,99	0,40	1,00	0,49	0,98	0,70	97,80	1,10	0,40	0,99
>0.3	PGTRW	1,00	0,67	1,00	0,49	1,00	0,56	1,00	0,10	99,60	0,10	0,10	1,00
	PG4000	1,00	0,67	1,00	0,51	1,00	0,58	1,00	0,10	99,70	0,10	0,10	1,00
	POTT1	1,00	0,63	1,00	0,44	1,00	0,52	1,00	0,10	99,60	0,20	0,10	1,00
	POTT2	1,00	0,63	1,00	0,42	1,00	0,51	1,00	0,10	99,60	0,20	0,10	1,00
	PLAMB	1,00	0,57	1,00	0,53	1,00	0,55	1,00	0,10	99,70	0,10	0,10	1,00

Accuracy (Accu): Measures how often the pluviometer correctly predicts rain/no rain.
Sensitivity (Recall) (Sens): Measures the pluviometer's ability to correctly identify rain events (true positive rate).
Specificity (Spec): Measures the pluviometer's ability to correctly identify non-rain events (true negative rate).
Positive Predictive Value (PPV): Measures the precision of rain predictions, indicating the proportion of predicted rain events that are actually rain events.
Negative Predictive Value (NPV): Measures the precision of no-rain predictions, indicating the proportion of predicted non-rain events that are actually non-rain events.
F1 Score (F1): A harmonic mean of precision (PPV) and recall (Sensitivity), balancing both false positives and false negatives.
Weighted F1 Score (F1w): A weighted version of the F1 score that accounts for the prevalence of different conditions (rain/no rain).
Percentage of True Positives (pTP): The proportion of actual rain events that are correctly identified as rain by the pluviometer.
Percentage of True Negatives (pTN): The proportion of actual non-rain events that are correctly identified as non-rain by the pluviometer.
Percentage of False Positives (pFP): The proportion of non-rain events that are incorrectly identified as rain by the pluviometer.
Percentage of False Negatives (pFN): The proportion of rain events that are incorrectly identified as non-rain by the pluviometer.
Cohen's Kappa (Kappa): Measures the agreement between the pluviometer and the reference sensor (PGPCP), accounting for the possibility of agreement occurring by chance.

Overall, from bin analysis we can appreciate considerable variability in the performance of different rain gauges compared with TBGPCP, depending on the threshold set for precipitation detection analysis. PGTRW tends to favor sensitivity over specificity, while POTT1 and POTT2 are more conservative, leading to higher specificity but lower sensitivity. PG4000 and PLAMB strike a more balanced approach, particularly at higher thresholds, making them potentially more reliable choices when considering both false positives and negatives.

Intercomparison : Maximum cross-correlations and lag.

The second step in the intercomparison is conducted to determine the cross-correlation and potential lags between the different pluviometers. The cross-correlation analysis compares all possible pairs of pluviometers at various aggregation intervals (1, 2, 4, 6, 8, 10, 20, 30 minutes). For each pair, cross-correlations are calculated across these intervals, along with the lag at which the maximum correlation occurs. In Figure 3, we can see that, for data at 1-minute resolution, a lag is present for -1 and 1 minute in 5 pair-cases, with maximum correlation ranging between 0.95 and 0.78. To evaluate performance at other aggregation times, we performed the same analysis for different temporal aggregations. The results indicate that the lag becomes 0 from 2 minutes of aggregation onwards, and cross-correlation increases rapidly from 2 minutes to 1 hour, reaching values over 0.98 for all pairs (not shown).

Intercomparison : multi-metric approach.

The third step in the intercomparison is to analyze and compare precipitation datasets using a multi-metric approach that employs various statistical, concordance, and dissimilarity metrics. This provides a comprehensive understanding of the similarities and differences in precipitation measurements among the pluviometers. In Table 5, we provide definitions for each metric used, with a focus on the less familiar ones: Jaccard, CCC, and ADI. The Jaccard Index is a measure of similarity between two sets, calculated as the intersection divided by the union of the sets of non-zero precipitation days. In this case, it captures the overlap of rainy days between two time series. The Concordance Correlation Coefficient (CCC) (Lin, 1989) is a valuable statistic for assessing agreement and reliability between two sets of continuous measurements. The Adaptive Dissimilarity Index (ADI) (Chouakria-Douzal and Nagabhushan, 2007) accounts for both dissimilarity on raw values and dissimilarity on temporal correlation behaviors.

Pair	Max Corr	Lag at Max Corr
PGPCP_PGTRW	0,88	-1
PGPCP_PG4000	0,87	0
PGPCP_POTT1	0,82	-1
PGPCP_POTT2	0,78	0
PGPCP_PLAMB	0,85	0
PGTRW_PG4000	0,95	1
PGTRW_POTT1	0,88	0
PGTRW_POTT2	0,84	1
PGTRW_PLAMB	0,94	1
PG4000_POTT1	0,89	0
PG4000_POTT2	0,85	0
PG4000_PLAMB	0,93	0
POTT1_POTT2	0,86	0
POTT1_PLAMB	0,85	0
POTT2_PLAMB	0,82	0

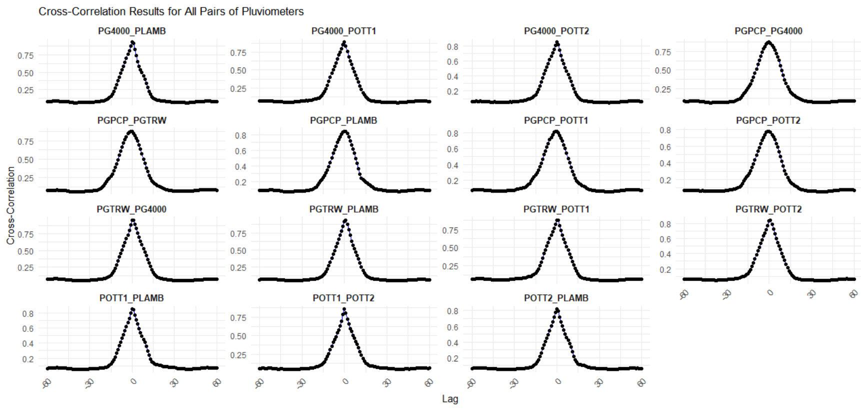

Figure 3: Results from cross-correlation betwen pluviometers pairs and cross-corr plot for selected episodes.

Table 5: Results from selected metrics in betwen pluvimeter pairs considering rain an no-rain events.

Pair	Jaccard	CCC	ADI	RMSE	rRMSE	MAE	nMAE	Pearson_Corr	Spearman_Corr	Max_Cor_Cross	Lead_Count	Lag_Count
PGPCP_PGTRW	0,58	0,88	0,26	0,018	2,16	0,005	0,57	0,91	0,74	0,91	920	3824
PGPCP_PG4000	0,57	0,88	0,25	0,019	2,22	0,005	0,64	0,90	0,70	0,90	2337	1422
PGPCP_POTT1	0,41	0,81	0,48	0,024	2,80	0,008	0,90	0,84	0,58	0,84	4180	413
PGPCP_POTT2	0,26	0,77	0,65	0,027	3,22	0,009	1,12	0,80	0,46	0,80	5374	213
PGPCP_PLAMB	0,56	0,86	0,47	0,020	2,34	0,006	0,71	0,87	0,68	0,87	1557	2942
PGTRW_PG4000	0,51	0,96	0,04	0,012	1,32	0,004	0,41	0,96	0,70	0,96	4651	843
PGTRW_POTT1	0,32	0,88	0,31	0,021	2,31	0,008	0,85	0,88	0,54	0,88	6889	189
PGTRW_POTT2	0,20	0,83	0,48	0,025	2,83	0,009	1,05	0,83	0,43	0,83	8189	88
PGTRW_PLAMB	0,57	0,92	0,28	0,016	1,83	0,005	0,51	0,92	0,71	0,93	3416	1866
PG4000_POTT1	0,39	0,89	0,29	0,020	2,18	0,008	0,83	0,89	0,53	0,89	3633	720
PG4000_POTT2	0,26	0,84	0,46	0,024	2,62	0,009	1,01	0,85	0,43	0,85	4631	360
PG4000_PLAMB	0,58	0,94	0,26	0,014	1,47	0,004	0,46	0,95	0,74	0,95	870	3196
POTT1_POTT2	0,42	0,85	0,18	0,024	2,70	0,007	0,82	0,85	0,58	0,85	1990	566
POTT1_PLAMB	0,34	0,85	0,16	0,023	2,56	0,009	0,97	0,85	0,51	0,85	394	5594
POTT2_PLAMB	0,21	0,82	0,28	0,026	2,91	0,010	1,13	0,82	0,41	0,82	214	6760

Jaccard Index: measures the similarity between two sets, specifically the intersection over the union of two datasets.
Concordance Correlation Coefficient (CCC): Assesses the degree to which pairs of observations fall on the 45-degree line through the origin, indicating perfect agreement.
Adaptive Dissimilarity Index (ADI): Measures dissimilarity between time series, considering both amplitude and phase differences.
Root Mean Square Error (RMSE) and Relative Root Mean Square Error (rRMSE): RMSE measures the average magnitude of errors between pairs of pluviometers, while rRMSE is the RMSE normalized by the mean.
Mean Absolute Error (MAE) and Normalized Mean Absolute Error (nMAE): Measure the absolute errors between pairs.
Pearson Correlation Coefficient (Pearson_Corr): This metric measures the linear correlation between two sets of data.
Spearman Correlation Coefficient (Spearman_Corr): This is a non-parametric measure of rank correlation, assessing how well the relationship between two variables can be described by a monotonic function.
Maximum Cross-Correlation (Max_Cor_Cross): This metric identifies the maximum correlation observed between time series at any lag.
Lead_Count and Lag_Count: These metrics indicate how often one pluviometer leads or lags another in detecting precipitation events.

Table 5 presents the results for all pluviometer pairs, with datasets filtered to exclude long periods with zero precipitation. The highest Jaccard Index is 0.55 for pairs PGPCP_PGTRW and PG4000_PLAMB, indicating these pairs have the most similar occurrences of rainfall events. The pair PGTRW_PG4000 shows the highest CCC of 0.96, with a narrow confidence interval, indicating strong agreement between these two pluviometers. The lowest ADI value is 0.04 for PGTRW_PG4000, suggesting that these two pluviometers have the most similar time series profiles. PGTRW_PG4000 also has the lowest RMSE and rRMSE, indicating the smallest average error between these pairs. PGTRW_PG4000 has the lowest MAE and nMAE, indicating minimal error normalized by the data range. The highest Pearson correlation is for PGTRW_PG4000 and PG4000_PLAMB, suggesting a strong linear relationship between these pluviometers. The highest Spearman correlation is 0.74 for PG4000_PLAMB, indicating a relatively strong monotonic relationship between these two pluviometers. PGTRW_PG4000 and PG4000_PLAMB have the highest maximum cross-correlation of 0.96 and 0.95, indicating strong alignment between these time series at some lag (see more in Figure 3). The highest lead count is 8189 for PGTRW_POTT2, and the highest lag count is 6760 for POTT2_PLAMB. These values suggest that PGTRW tends to detect precipitation before POTT2, and POTT2 tends to lag behind PLAMB.

The most concordant pairs are PGTRW_PG4000 and PG4000_PLAMB, based on several metrics. The PGTRW_PG4000 pair has the highest CCC, Pearson correlation, and maximum cross-correlation. It also has the lowest ADI, indicating very high similarity in both amplitude and phase. These metrics collectively indicate that these two pluviometers are highly concordant in their measurements. The PG4000_PLAMB pair shows a high CCC (0.93), Pearson correlation (0.93), and maximum cross-correlation (0.93). It also has the lowest RMSE (0.00), suggesting minimal errors between these time series. These values indicate strong concordance between PG4000 and PLAMB.

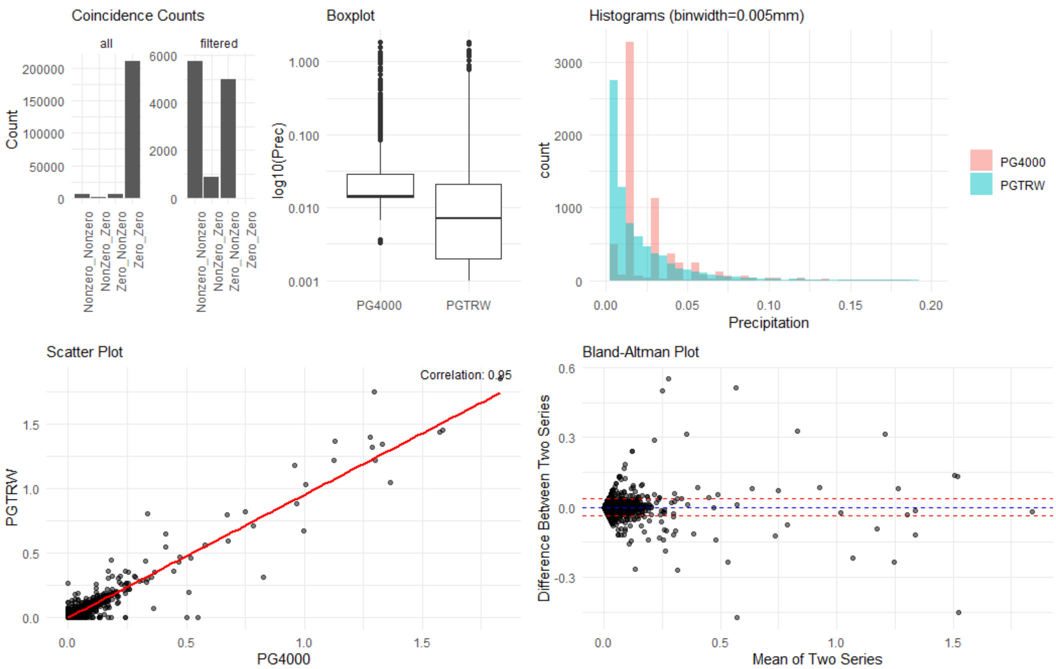

Figure 4: Example of some graphs used for visual characterization of the relationship in betwen two pluviometric data series.

Intercomparison: graphs

In Figure 4, we present examples of graphical representations used for visual analysis of the pairs of series. As a graphical summary of the most relevant aspects of the interactions between both series, we include bar plots, box plots, histograms, scatter plots, and Bland-Altman plots. The Bar Plot of Coincidences visualizes the counts of different combinations: Nonzero_Nonzero (both series have a non-zero value), Zero_Zero (both series have a zero value), Zero_NonZero (the first series is zero and the second series is non-zero), and NonZero_Zero (the first series is non-zero and the second series is zero). This bar graph provides a quick overview of how often both series have zero values or differ in terms of zero and non-zero values for all data, including data filtered by removing coincident zeros. The boxplot provides a visual comparison of the median (central line), interquartile range (IQR), and potential outliers. The histogram displays the count distributions of precipitation values for the two series, allowing comparison of their overall distribution by bins of 0.005 mm. A scatter plot with correlation shows the relationship between the two series with a regression line, along with a text annotation indicating the Pearson correlation coefficient. A Bland-Altman Plot shows the agreement between the two series, with mean difference and limits of agreement (±1.96 standard deviations).

Efects of temporal agregation

o check the effects of aggregation time intervals, different studies are carried out to characterize each dataset at different temporal aggregations and to assess the impact of aggregation on the quality of pair metrics. For this purpose, data at 1 minute are aggregated to 2, 4, 6, 8, 10, 20, 30, 40, 50 minutes, and to 1, 2, 3, 6, 12, and 24 hours. After all, some statistics are extracted for comparison purposes.

In Table 3, we present an example with different statistics and counts for four different temporal resolutions. Mean values for all aggregated pluviometers show logical consequences of aggregation. We observe fewer events, lower maximum precipitation, fewer zero periods, and other obvious consequences. Additionally, lag and lead counts decrease as the aggregation interval increases. All intercomparison metrics improve as the aggregation time increases (not shown).

Table 3: Effects of temporal agregation.

	Max_NonZero	SD_NonZero	Rain_Events	NoRain_Events	NoZero_Periods	Zero_Periods	Total_Accumulated_Precip	Avg_Accumulated_Precip	Max_Accumulated_Precip	Avg_Rain_Duration	Max_Rain_Duration	Min_Rain_Duration	Avg_NonRain_Duration	Max_NonRain_Duration	Min_NonRain_Duration	Lead_Count_PGPCP	Lag_Count_PGPCP
1min	1,7	0,1	6721	216482	560	923	210	0,2	20,1	7,4	140	2,5	231	15200	2,5	2000	3177
10min	13,5	0,5	1271	21052	199	236	210	0,8	20,9	44	382	11,7	775	15217	10,0	520	256
1hour	16,3	1,1	392	3331	78	96	210	1,7	26,7	188	1620	60,0	1655	15170	60,0	192	91
24hour	24,5	4,5	76	82	16	19	210	6,7	40,4	3849	13680	1440,0	4037	13920	1440,0	39	22

4. Conclusions

Rainfall data collected at one-minute intervals from five distinct weighing gauges (Geonica, OTT, Lambrecht) and an operational tipping-bucket rain gauge (10 min) located at the same site are evaluated and intercompared during a continuous five-month period and two different episodes.

A 2x3 (2 stages x 3 steps each) methodology has been developed, proving effective for the evaluation and intercomparison of the different instruments in our case. This methodology can serve as a basis for other studies of the same type or apply some of the methodological conclusions drawn to different phases of data quality assurance processes (Hernandez 2011, 2012).

A careful selection of more or less complex metrics has been made to evaluate and intercompare different aspects of the datasets globally and considering them as time series. Different scripts have been implemented in R in all cases, enabling efficient and recursive calculations.

Maximum cross-correlation and lag analysis seems very useful for checking temporal inconsistencies in data acquisition. Period-based analysis provides a good framework for characterizing different series. In the future, it would be interesting to apply these techniques to aggregated precipitation episodes.

Currently, a disdrometer is installed at the site, so it will be included in the comparative study in the near future to gain insights into specific aspects of interest, such as the behavior of these instruments under different types of precipitation (drizzle, rain, snow, or hail).

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6. References

Campbel scientific. (2013). CR1000 Operators Manual.

Colotti E., Blanco A., Rodríguez J.F42 (2003). Aplicación del coeficiente de inconsistencia como criterio de comparación entre mediciones Convencionales y Automáticas de lluvia diaria. Terra Nueva Etapa. 19.

Chouakria, A. D., & Nagabhushan, P. N. (2007). Adaptive dissimilarity index for measuring time series proximity. Advances in Data Analysis and Classification, 1(1), 5–21. http://doi.org/10.1007/s11634-006-0004-6

Duchon, C. E. and Biddle, C. J. (2010) Undercatch of tipping-bucket gauges in high rain rate events , Advances in Geosciences, 25, pp. 11–15

Euskalmet (2022). Basque Meteorology Agency Web. https://www.euskalmet.euskadi.eus/inicio/

Gaztelumendi, S., and Aranda, J. A. Comparative study of rain measurement by optical disdrometers and tipping-bucket rain gauges in Basque Country. WMO Technical Conference on Meteorological and Environmental Instruments and Methods of Observation (TECO-2024) 23-26 September 2024, Vienna (Austria). 2024a.

Gaztelumendi, S., Egaña, J., and Aranda, J. A.: On the use of disdrometer data for characterization of precipitation episodes in the Basque Country, EMS Annual Meeting 2024, Barcelona, Spain, 1–6 Sep 2024, EMS2024-454, https://doi.org/10.5194/ems2024-454, 2024b.

Gaztelumendi S, Gil-Garcia JM, Otxoa de Alda K., Maiz A, Olabe A, Maruri M, Aranda J.A. (2022a) Modernisation of the Basque AWS network : rain gauges. Proceedings 2022 WMO/CIMO Technical Conference on Meteorological and Environmental Instruments and Methods of Observation (CIMO TECO-2022).

Gaztelumendi S, Otxoa de Alda K., Ruiz, R.,Orue J., Egaña J., Maruri M., Aranda J.A. (2022b). High temporal resolution monitoring of precipitation in the Basque Country. European Meteorology Conference EMS Annual Meeting 2022, 4–9 September, Bohn- Online

Gaztelumendi S, Otxoa de Alda K., Gomez de Segura JD, Orue J., Aranda J.A. (2022c). On the operational use of 1-min rain data in Basque Meteorology Agency. Proceedings 2022 WMO/CIMO Technical Conference on Meteorological and Environmental Instruments and Methods of Observation (CIMO TECO-2022).

Gaztelumendi S., Otxoa de Alda, K., R. Hernández, R., M. Maruri, Aranda, J. A. and Anitua, P. (2018). The Basque Automatic Weather Station Mesonetwork in perspective. Proceedings 2018 WMO/CIMO Technical Conference on Meteorological and Environmental Instruments and Methods of Observation (CIMO TECO-2018).

Gaztelumendi, S., Otxoa de Alda, K., and Hernandez, R. (2003) Some aspects on the operative use of the automatic stations network of the Basque country, 3th ICEAWS Conference, 2003.

Hernández R, M. Maruri, K. Otxoa de Alda, J. Egaña, S. Gaztelumendi. (2011) Quality Assurance in the Basque Country AWS Mesonet. 11th EMS Annual Meeting (EMS) 10th European Conference on Applications of Meteorology (ECAM) Berlin, (Germany), 12 – 16 September 2011.

Hernández R., Maruri, K. Otxoa de Alda, J. Egaña, and S. Gaztelumendi (2012) Quality control procedures at Euskalmet data center .Adv. Sci. Res., 8, 129-134, https://doi.org/10.5194/asr-8-129-2012, 2012.

Liu, X. C., Gao, T. C., and Liu, L. (2013) A comparison of rainfall measurements from multiple instruments, Atmos. Meas. Tech., 6, 1585–1595. 2013.

Lin LI. A concordance correlation coefficient to evaluate reproducibility. Biometrics. 1989 Mar;45(1):255-68. PMID: 2720055.

Rodda, J. C., & Dixon, H. (2012). Rainfall measurement revisited. Weather, 67, 131– 136.

Sevruk, B., Ondrás, M. and Chvíla, B. (2009) 'The WMO precipitation measurement intercomparisons', Atmospheric Research. Elsevier B.V., 92(3), pp. 376–380. doi: 10.1016/j.atmosres.2009.01.016.

WMO (2009) Field Intercomparison of rainfall intensity gauges. Instruments and Observing Methods Report No. 99, WMO Geneva, Switzerland: WMO/TD-No. 1504.

WMO (2018): World Meteorological Organization Guide to Meteorological Instruments and Methods of Observation, WMO-No. 8, Geneva. 2018.